João Augusto Lopes

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Profile

Robotics Engineering graduate with a strong multidisciplinary background, with specialisation on vision-driven robotic control and real-time perception. Passionate about Artificial Intelligence and Autonomous Systems. Highly motivated and continuously striving to learn and develop new skills. Eager to apply academic knowledge and practical experience to contribute to advanced research and knowledge development in the field.

Publications

D. Vergara, S. Shabana, **J. Lopes**, M. S. Erden, and M. Koskinopoulou, "Ultrasound-guided Robotic Aspirator for Autonomous Thoracentesis in Pleural Effusion Treatment," in *2025 International Conference on Robotics and Mechatronics (ICARM)*, to appear, 2025.

Co-designed and implemented a robotic system, using ROS and a robotic manipulator, for autonomous ultrasound-guided thoracentesis. Developed a C++ vision framework combining binary masking, morphological operations, Canny filtering, and Hough Transforms for pleural zone detection (95.6% accuracy), with ORB for needle tracking and real-time feedback. Integrated embedded actuators and ROS-Arduino control for autonomous needle insertion and fluid aspiration. Informed system design through a thorough investigation of prior work, ensuring relevance to current research directions. Additionally, contributed to the preparation and writing of the associated publication.s

Education

Sep 2020 – Jun 2025

Masters of Engineering with First Class Honours: Robotics Autonomous and Interactive Systems

Heriot-Watt University, Edinburgh

The course covered a broad range of topics including electronics, computer science and mechanics, providing a comprehensive understanding of autonomous systems and their real-world applications. Through hands-on projects and research collaborations, I developed both the technical expertise and practical skills required for innovative developments in robotics and intelligent systems

Relevant Research:

Imitation Learning for Robotic Tracking of Human Wrist Trajectories

MEng 5th Year Research Placement

Developed a real-time robotic system for wrist trajectory imitation using a robotic manipulator, an RGB-D camera and ROS. Implemented a vision pipeline in C++ for wrist detection through colour segmentation, contour analysis, and 3D projection. Created an algorithm that uses the wrist's 3D position to control the robotic arm's end-effector in real time. Conducted a comprehensive literature review of state-of-the-art methods to inform system design and guide implementation choices. Also authored a thorough technical report detailing system architecture, implementation challenges, and analysis of results.

Key Modules:

- Signals and Signals

    - Analysed continuous-time signals and systems using time-domain methods (impulse response, convolution) and frequency-domain techniques (Fourier and Laplace transforms). Modelled and simulated system behaviour in RLC circuits using MATLAB.
- Time Frequency and Signal
    - Implemented signal processing techniques, in MATLAB, including FFT spectral analysis, modulation (AM, FM, QAM), and DTMF decoding, with focus on sampling theory and windowing effects.
- *Machine Vision*
    - Explored visual language concepts, in MATLAB, through image classification using SURF feature extraction and Bag of Visual Words representations, followed by k-means clustering and SVM classification. Compared it with a fine-tuned ResNet-18 CNN on CIFAR-10 dataset.
    - Created an algorithm, in MATLAB, for 3D Image Reconstruction using Epipolar Geometry.
- *Biologically Inspired Computation*
    - Developed an Artificial Neural Network from scratch, in Python, with Particle Swarm Optimisation for Supervised Learning.
- *Image Processing*
    - Analysed images, in MATLAB, using skeletonization and Fourier descriptors to extract key features and ensure invariance to transformations.
- *Artificial Intelligence and Intelligent Agents*
    - Built an intelligent game, in Python, using Reinforcement Learning with Markov Decision Process.
    - Applied NLP techniques including n-gram models, syntactic parsing, and grammar formalism (Chomsky Normal Form) in practical contexts.
- *Intelligent Robotics*
    - Developed a navigation system, in Python, using both behaviour-based robotics and evolutionary robotics approaches.

Sep 2016 – Jul 2020
European Baccalaureate: Physics and Mathematics
European School of Brussels II, Brussels

Positions of Responsibility

Sep 2023 – Jul 2025
Tournament Director
Heriot-Watt Ultimate Frisbee Club
Organised very successful tournaments with over 200 athletes. Also participated in club management activities, providing support to the committee in various administrative tasks.

Achievements

Mar 2019
3rd place at European Schools Science Symposium 2019
Built an Arduino based device, "Air Pollution Detector", that detects harming gases in the atmosphere near you and sends a text message to you when the gases concentrations reach dangerous values.